Practice Exam - 1

## Question 74 of 117

A 61-year-old man develops cardiogenic shock with severe hypotension complicating acute myocardial infarction (MI) necessitating cannulation on femoral-femoral peripheral veno arterial extracorporeal membrane oxygenation (VA-ECMO). His course was also complicated by an aspiration event with subsequent pneumonia and acute respiratory distress syndrome (ARDS) with severe hypoxemia. The day after cannulation, he has improving but still depressed heart function with aortic valve ejection while supported with low dose epinephrine at 2 mcg/min and full VA ECMO support with flows of 6.5 L/min. He has persistent severe ARDS with minimal native lung function. His ventilator is set to AC VC setting with an Fio2 of 80% and PEEP of 5mmHg. Pulse oximetry on the right hand reads 78%, and a right radial arterial blood gas shows PaO2 of 48 mmHg, while a femoral arterial line distal to the ECMO cannula shows PaO2 of 210 mmHg.

What is the most appropriate next step in management?

**A.** Increase femoral arterial ECMO flow only

**B.** Convert to veno-venous (VV) ECMO configuration

**C.** Initiate a beta-blocker to suppress residual native cardiac output

**D.** Add a second arterial return cannula (e.g., right axillary) to create a veno-arterial-arterial (VAV) ECMO configuration

**E.** Increase peep on the ventilator and continue current cannulation

Commentary | References

**Content Code (Blueprint):** Critical Care Cardiology — Mechanical Circulatory Support Complications (AHFTC Blueprint)

**Task:** Diagnosis

**Teaching Point:** On peripheral veno arterial extracorporeal membrane oxygenation (VA-ECMO) with recovering native cardiac function but persistent severe lung injury, oxygenated ECMO blood perfuses the lower body while poorly oxygenated native cardiac output preferentially perfuses the upper body/brain, producing differential hypoxemia known as North-South (Harlequin) syndrome.

**Rationale:**

Correct answer: D. Add a second arterial return cannula (e.g., right axillary) to create a veno-arterial-arterial (VAV) ECMO configuration

VAV ECMO: As native cardiac ejection recovers, the heart pumps poorly oxygenated blood from the failing lungs forward into the aorta, mixing antegrade with the retrograde, well-oxygenated ECMO flow at a point determined by cardiac output and ECMO flow (the 'mixing cup'). Adding a second arterial return cannula (often internal jugular) delivers oxygenated blood to the upper body/coronaries/brain, correcting the differential hypoxemia.

**Incorrect answers:**

A. Increasing femoral ECMO flow alone: This shifts the mixing point cephalad but does not reliably correct upper-body hypoxemia and may increase LV afterload and risk LV distention. Further, the ECMO flows are already relatively high and further increases would likely not be effective and could risk complication such as hemolysis, thrombocytopenia etc.

B. Converting to VV-ECMO: This is inappropriate while the patient still has significant cardiac dysfunction requiring hemodynamic support as this configuration provides no circulatory support.

C. Beta-blockade to suppress native output: Pharmacologically inducing negative inotropy in a patient with cardiogenic shock, especially one requiring epinephrine support is counter-productive and is not a standard or safe strategy and ignores the underlying respiratory failure.

D. Correct answer

E. Increasing PEEP: Native lung disease (e.g., severe ARDS) limits oxygen transfer regardless of PEEP.

**Correct Answer (Best Answer Choice):** Add a second arterial return cannula (e.g., right internal jugular) to create a veno-arterial-arterial (VAV) ECMO configuration

**Distractors (Plausible Incorrect Options):**

- Increase femoral arterial ECMO flow only
- Convert to veno-venous (VV) ECMO configuration
- Initiate a beta-blocker to suppress residual native cardiac output
- Increase FiO2 on the ventilator alone and continue current cannulation